

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250284001

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Ching; Sze Ping et al.

SYSTEMS AND METHODS TO DETECT IMPROPER MEDIA LOADING

Abstract

A system to detect improper media loading is disclosed. The system comprises at least one media sensor installed within a printer and configured to determine information related to an angle of at least one arm that rotates in a predefined path upon receiving a media within printer, or a distance between media and at least one media sensor. Further, at least one processor is communicatively coupled to the at least one media sensor and configured to receive determined information related to angle or distance between media and at least one media sensor, from the at least one media sensor. The at least one processor compares the determined information with a predefined threshold value range associated with angle of at least one arm or distance. Thereafter, the at least one processor determines improper loading of the media within the printer based at least on the comparison.

Inventors: Ching; Sze Ping (Singapore, SG), Ng; Cheng Khoon (Singapore, SG), Lee; Phek Thong (Singapore, SG)

Applicant: Hand Held Products, Inc. (Charlotte, NC)

Family ID: 96949044

Appl. No.: 18/598578

Filed: March 07, 2024

Publication Classification

Int. Cl.: G01S15/08 (20060101); G01S15/86 (20200101); H04N1/00 (20060101)

U.S. Cl.:

CPC G01S15/08 (20130101); G01S15/86 (20200101); H04N1/00689 (20130101);
H04N1/00694 (20130101);

Background/Summary

TECHNOLOGICAL FIELD

[0001] Example embodiments of the present disclosure relate generally to printers, and more particularly, to a system and method to detect improper media loading in a printer.

BACKGROUND

[0002] Printers play an indispensable role in facilitating efficient document processing. Lack of a reliable system to confirm proper media loading within a printer is a critical issue existing in the printers. Moreover, users are left without any feedback mechanism to ascertain whether the media, such as paper or other printing materials, is loaded correctly into the printer. Absence of such feedback mechanism has significant consequences, ranging from inconvenient paper jams to the impairment of essential media low detection features. Therefore, users are left to navigate through the process of improper media loading without real-time confirmation, leading to a lot of issues. Also, the improper loading of media is a primary cause behind frequent paper jams that disrupts workflow. Such improper loading of media not only disrupts printing tasks but also risks damage to the hardware of the printer and compromises the overall user experience.

[0003] The inventors have identified numerous areas of improvement in the existing technologies and processes, which are the subjects of embodiments described herein. Through applied effort, ingenuity, and innovation, many of these deficiencies, challenges, and problems have been solved by developing solutions that are included in embodiments of the present disclosure, some examples of which are described in detail herein.

BRIEF SUMMARY OF THE INVENTION

[0004] The following presents a summary of some example embodiments to provide a basic understanding of some aspects of the present disclosure. This summary is not an extensive overview and is intended to neither identify key or critical elements nor delineate the scope of such elements. It will also be appreciated that the scope of the disclosure encompasses many potential embodiments in addition to those here summarized, some of which will be further described in the detailed description that is presented later.

[0005] In an example embodiment, a system is disclosed. The system comprises at least one media sensor installed within a printer. The at least one media sensor is configured to determine information related to an angle of at least one arm that rotates in a predefined path upon receiving a media within the printer, or a distance between the media and the at least one media sensor. Further, at least one processor is communicatively coupled to the at least one media sensor. The at least one processor is configured to receive the determined information related to the angle or the distance between the media and the at least one media sensor, from the at least one media sensor. Further, the at least one processor is configured to compare the determined information with a predefined threshold value range associated with the angle of the at least one arm or the distance between the media and the at least one media sensor. Thereafter, the at least one processor is configured to determine improper loading of the media within the printer based at least on the comparison.

[0006] In some embodiments, the at least one media sensor comprises at least one of an angular sensor or a proximity sensor. In some embodiments, the angular sensor has the at least one arm, and is configured to determine the angle of the at least one arm. In some embodiments, the proximity sensor emits one or more rays and receives reflected one or more rays from the media to determine the distance between the proximity sensor and the media. In some embodiments, the one or more rays corresponds to at least one of one or more infrared waves (IR) or ultrasonic sound waves (UV).

[0007] In some embodiments, the at least one arm of the angular sensor is positioned along a surface of the media, and the proximity sensor is positioned at a base of the printer and underneath

to the media. The predefined path of the at least one arm defines a first end and a second end. In some embodiments, the at least one arm positioned at the first end of the predefined path indicates an empty media condition and the at least one arm positioned at the second end of the predefined path indicates a full media condition.

[0008] In some embodiments, the media pushes the at least one arm towards the second end of the predefined path that corresponds to the proper loading of the media within the printer. In some embodiments, the determined angle of the at least one arm below or above the predefined threshold value range corresponds to the improper loading of the media within the printer. In some embodiments, the determined distance between the media and the proximity sensor below or above the predefined threshold value range corresponds to the improper loading of the media within the printer.

[0009] In some embodiments, the at least one processor is configured to generate one or more notifications upon determining the improper loading of the media within the printer. In some embodiments, the one or more notifications are displayed to a user over a display device communicatively coupled to the at least one processor.

[0010] In another example embodiment, a method is disclosed. The method comprises steps of determining, via at least one media sensor installed within a printer, information related to an angle of at least one arm that rotates in a predefined path upon receiving a media within the printer, or a distance between the media and the at least one media sensor; receiving, via at least one processor, the determined information related to the angle or the distance between the media and the at least one media sensor, from the at least one media sensor; comparing, via the at least one processor, the determined information with a predefined threshold value range associated with the angle of the at least one arm or the distance between the media and the at least one media sensor; and determining, via the at least one processor, improper loading of the media within the printer based at least on the comparison.

[0011] The above summary is provided merely for purposes of summarizing some example embodiments to provide a basic understanding of some aspects of the present disclosure.

Accordingly, it will be appreciated that the above-described embodiments are merely examples and should not be construed to narrow the scope or spirit of the present disclosure in any way. It will be appreciated that the scope of the present disclosure encompasses many potential embodiments in addition to those here summarized, some of which will be further described below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Having thus described certain example embodiments of the present disclosure in general terms, reference will hereinafter be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0013] FIG. 1 illustrates a block diagram showing a system to detect improper loading of a media in a printer in accordance with an example embodiment of the present disclosure;

[0014] FIGS. 2A-2B illustrate a media holder of a printer in accordance with an example embodiment of the present disclosure;

[0015] FIG. 2C illustrates a full media condition detected by an angular sensor in accordance with an example embodiment of the present disclosure;

[0016] FIGS. 2D-2E illustrate a media hanger at each side of the media and each media hanger having the angular sensor in accordance with an example embodiment of the present disclosure;

[0017] FIG. 3 illustrates a proximity sensor integrated within the media holder of the printer in accordance with an example embodiment of the present disclosure;

[0018] FIG. 4 illustrates a display device of the system in accordance with an example embodiment

of the present disclosure; and

[0019] FIG. 5 illustrates a flowchart showing steps of a method to detect improper loading of a media in accordance with an example embodiment of the present disclosure.

DETAILED DESCRIPTION

[0020] Some embodiments will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all, embodiments of the present disclosure are shown. Indeed, various embodiments may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements.

[0021] The components illustrated in the figures represent components that may or may not be present in various embodiments of the present disclosure described herein such that embodiments may include fewer or more components than those shown in the figures while not departing from the scope of the present disclosure. Some components may be omitted from one or more figures or shown in dashed line for visibility of the underlying components.

[0022] As used herein, the term “comprising” means including but not limited to and should be interpreted in the manner it is typically used in the patent context. Use of broader terms such as comprises, includes, and having should be understood to provide support for narrower terms such as consisting of, consisting essentially of, and comprised substantially of.

[0023] The phrases “in various embodiments,” “in one embodiment,” “according to one embodiment,” “in some embodiments,” and the like generally mean that the particular feature, structure, or characteristic following the phrase may be included in at least one embodiment of the present disclosure and may be included in more than one embodiment of the present disclosure (importantly, such phrases do not necessarily refer to the same embodiment).

[0024] The word “example” or “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations.

[0025] If the specification states a component or feature “may,” “can,” “could,” “should,” “would,” “preferably,” “possibly,” “typically,” “optionally,” “for example,” “often,” or “might” (or other such language) be included or have a characteristic, that a specific component or feature is not required to be included or to have the characteristic. Such a component or feature may be optionally included in some embodiments or it may be excluded.

[0026] The present disclosure provides various embodiments of systems and methods to detect improper loading of media. Embodiments may be configured to determine improper loading of a media within a media holder of the printer. Embodiments may be configured to allow movement of at least one arm of an angular sensor to move from one end to other end upon loading or unloading of the media within the media holder. Embodiments may be configured to determine an angle of the at least one arm of an angular sensor upon loading of the media within the media holder. Embodiments may be configured to determine distance between the media and a proximity sensor installed within the media holder upon loading of the media within the media holder. Embodiments may be configured to compare the determine angle and the distance with a predefined threshold value range to determine improper loading or unloading of the media within the media holder.

[0027] FIG. 1 illustrates a block diagram showing a system **100** to detect improper loading of media in a printer **102**, in accordance with an example embodiment of the present disclosure. The system **100** may comprise at least one media sensor **104**, at least one processor **106**, a memory **108**, and a display device **110**.

[0028] In some embodiments, the at least one media sensor **104** may be installed within the printer **102**. Further, the at least one media sensor **104** may comprise at least one of an angular sensor **112** having at least one arm (not shown) or a proximity sensor **114**. Further, the at least one media sensor **104** may be configured to determine information related to improper media loading within the printer **102**. In some example embodiments, the angular sensor **112** may be configured to

determine the angle of the at least one arm that may rotate in the predefined path. Further, the at least one media sensor **104** may be configured to determine a distance between the media (not shown) and the at least one media sensor **104**. In some other example embodiments, the proximity sensor **114** may be configured to determine the distance between the media and the proximity sensor **114**. The detailed description of the at least one media sensor **104** is described in conjunction with FIGS. 2A-3.

[0029] As illustrated in FIG. 1, the system **100** may further comprise the at least one processor **106**. The at least one processor **106** may be communicatively coupled to the at least one media sensor **104**. Further, the at least one processor **106** may be configured to receive the determined information related to the angle or the distance between the media and the at least one media sensor **104**, from the at least one media sensor **104**. Further, the at least one processor **106** may be configured to compare the determined information with a predefined threshold value range associated with the angle of the at least one arm or the distance between the media and the at least one media sensor **104**. Further, the at least one processor **106** may be configured to determine improper loading of the media within the printer **102** based at least on the comparison.

[0030] In various example embodiments, the at least one processor **106** may include suitable logic, circuitry, and/or interfaces that are operable to execute one or more instructions stored in the memory **108** to perform predetermined operations. In some embodiments, the at least one processor **106** may be configured to decode and execute any instructions received from one or more other electronic devices or server(s). The at least one processor **106** may be configured to execute one or more computer-readable program instructions, such as program instructions to carry out any of the functions described in this description. Further, the at least one processor **106** may be implemented using one or more processor technologies known in the art. Examples of the processor include, but are not limited to, one or more general purpose processors (e.g., INTEL® or Advanced Micro Devices® (AMD) microprocessors) and/or one or more special purpose processors (e.g., digital signal processors or Xilinx® System On Chip (SOC) Field Programmable Gate Array (FPGA) processor).

[0031] Further, the memory **108** may store a set of instructions and data. In some embodiments, the memory **108** may include the one or more instructions that are executable by the at least one processor **106** to perform specific operations. In some embodiments, the memory **108** may be configured to store the received determined information related to the angle or the distance between the media and the at least one media sensor **104**, from the at least one media sensor **104**. Further, the memory **108** may be configured to store the compared values of the determined information with the predefined threshold value range associated with the angle of the at least one arm or the distance between the media and the at least one media sensor **104**. It is apparent to a person with ordinary skill in the art that the one or more instructions stored in the memory **108** enable the hardware of the system **100** to perform the predetermined operations. Some of the commonly known memory implementations include, but are not limited to, fixed (hard) drives, magnetic tape, floppy diskettes, optical disks, Compact Disc Read-Only Memories (CD-ROMs), and magneto-optical disks, semiconductor memories, such as ROMs, Random Access Memories (RAMs), Programmable Read-Only Memories (PROMs), Erasable PROMs (EPROMs), Electrically Erasable PROMs (EEPROMs), flash memory, magnetic or optical cards, or other type of media/machine-readable medium suitable for storing electronic instructions.

[0032] In some embodiments, the at least one processor **106** may be configured to generate one or more signals to cause one or more notifications upon determining the improper loading of the media **210** (FIG. 2B) within the printer **102**. Further, the one or more notifications may be displayed for a user over the display device **110**. The display device **110** may be communicatively coupled to the at least one processor **106**. The display device **110** is described in greater detail in conjunction with FIG. 4.

[0033] It will be apparent that the above-mentioned components of the system **100** have been

provided only for illustration purposes. In another embodiment, the system **100** may include other components such as a controller unit, a microprocessor unit (MPU), a microcontroller unit (MCU), etc. without departing from the scope of the disclosure.

[0034] FIGS. **2A-2B** illustrate at least one media holder **200** of the printer **102**, in accordance with an example embodiment of the present disclosure. FIG. **2C** illustrates a full media condition detected by the angular sensor **112**, in accordance with an example embodiment of the present disclosure. FIGS. **2A-2C** are described in conjunction with FIG. **1**.

[0035] The at least one media holder **200** may ensure proper alignment of the media **210** as the media **210** passes through various components (not shown) of the printer **102**. Further, the at least one media holder **200** may ensure smooth movement of the media **210** as the media **210** passes through the various components of the printer **102**. In some embodiments, the at least one media holder **200** may comprise the angular sensor **112**. Further, the angular sensor **112** may include at least one arm **202**. The at least one arm **202** of the angular sensor **112** may be positioned along a surface of the media **210**. As discussed earlier, the at least one media sensor **104**, e.g., the angular sensor **112**, may be configured to determine information related to the angle of the at least one arm **202** that rotates in a predefined path **204** upon receiving the media **210** within the printer **102**.

Further, the angular sensor **112** may measure angular rotation of the at least one arm **202**.

Thereafter, the angular sensor **112** may convert the measured angular rotation into a scaled electrical signal in real time to determine information related to the angle of the at least one arm **202**. In some embodiments, the media **210** may include printing media such as paper, labels, or other printable materials known in the art.

[0036] Further, the predefined path **204** of the at least one arm **202** may define a first end **206** and a second end **208**. The at least one arm **202** may be positioned between the first end **206** and the second end **208**. In some embodiments, the at least one arm **202** positioned at the first end **206** may indicate an empty media condition. In some example embodiments, the empty media condition may correspond to a condition where the media of the printer **102** is completely used. In some other embodiments, the at least one arm **202** positioned at the second end **208** of the predefined path **204** may indicate a full media condition. In some example embodiments, the full media condition may correspond to a condition where a new media is loaded within the printer **102**.

[0037] In some embodiments, while positioning the at least one arm **202** between the first end **206** and the second end **208** and along the predetermined path **204**, the angle of the at least one arm **202** may be determined with respect to a horizontally-extending axis A that the at least one arm **202** rotates around. Further, the angle of the at least one arm **202** may be directly proportional to diameter of the media **210**, e.g., the angle may change when the diameter of the media may change. In some embodiments, the diameter of the media **210** may also correspond to amount of media remaining on the media hanger, e.g. as the media **210** gets used by the user, the diameter of the media **210** may also reduce. In some embodiments, improper loading of media may be detected based on a predefined threshold value range of the angle. In some embodiments, the predefined threshold value range may depend upon diameter of the media in order to convert the electrical signal reading to the diameter of the media. Further, if the electrical signal reading value is below or above the predefined threshold value range or the diameter of the media, then improper loading of the media may be indicated. In one example embodiment, the predefined threshold value range may include a range of 0° to 1°. Further, the predefined threshold range value may depend upon the design of the at least one arm **202** of the angular sensor **112**. In some example embodiments, the determined angle of the at least one arm **202** below or above the predefined threshold value range may correspond to the improper loading of the media within the printer **102**.

[0038] In some embodiments, a media **210** as shown in FIG. **2B**, may be placed within the at least one media holder **200** of the printer **102**. Further, the media **210** may be resting over a media hanger. The media hanger may comprise one or more adjustable brackets or one or more arms that may be designed to secure the media **210** in a fixed position during the printing operation. The

media **210** when placed within the at least one media holder **200**, may press the at least one arm **202**. The pressing of the at least one arm **202** may reposition the at least one arm **202** from the first end **206** of the predefined path **204** to the second end **208** of the predefined path **204**. In some embodiments, upon removing the media **210** from the media holder, the at least one arm **202** may retract back from the second end **208** of the predefined path **204** to the first end **206** of the predefined path **204**. In some exemplary embodiments, the at least one arm **202** may be integrated with a biasing member, such as a spring (not shown). The biasing member may be configured to retract the at least one arm **202** from the second end **208** of the predefined path **204** to the first end **206** of the predefined path **204** upon unloading the media **210**. In some embodiments, the determined angle of the at least one arm **202** below or above the predefined threshold value range may correspond to the improper loading of the media **210** within the printer **102**.

[0039] FIGS. 2D-2E illustrate a portion of a printer **102**, in accordance with an example embodiment of the present disclosure. FIGS. 2D-2E are described in conjunction with FIGS. 2A-2C.

[0040] In some embodiments, each side of the media **210** may be installed with the media hanger. Further, the printer **102** may include two or more angular sensors **112**, at least one angular sensor **112** may be positioned on an opposite internal side of the printer **102** as at least one other angular sensor **112**. In some embodiments, each angular sensor **112** may include at least one arm **202** positioned along the surface of the media **210**. Each angular sensor **112** may be configured to determine information related to the angle of the at least one arm **202** that rotates in the respective predefined path **204** upon receiving the media **210** within the printer **102**. Further, each angular sensor **112** may measure angular rotation of the at least one arm **202**. In some embodiments, upon placing the media **210** within the media hanger, each arm of the angular **112** sensor at each side of the media may rotate in the respective predefined path **204** as shown in FIG. 2E.

[0041] Further, the predefined path **204** of each arm **202** may define the first end **206** and the second end **208**. Each arm **202** may be positioned between the respective first end **206** and the second end **208**. In some embodiments, each arm **202** positioned at the first end **206** may indicate an empty media condition that correspond to the condition where the media of the printer **102** is completely used. In some other embodiments, each arm **202** positioned at the second end **208** of the predefined path **204** may indicate the full media condition, which may correspond to the condition where the new media is loaded within the printer **102**.

[0042] In some embodiments, improper loading of media **210** may be detected based on comparing the angle of rotation of the at least one arm **202** of the angular sensor **112** at each side of the media **210**. For example, improper loading of media **210** may be detected by comparing the angle of rotation of at least one angular sensor **112** that is positioned on a first internal side of the printer **102** to the angle of rotation of at least one other angular sensor **112** that is positioned on an opposite second internal side of the printer **102**. In some embodiments, any difference between the angle of rotation of the at least one arm **202** of the angular sensor **112** at each side of the media may correspond to improper loading of the media **210**. In some embodiments, the difference between the angle of rotation of each of the at least one arm **202** above a threshold value may correspond to the improper loading of the media **210**. Further, the difference between the angle of rotation of each of the at least one arm **202** below a threshold value may correspond to the proper loading of the media **210**. In one example embodiment, the predefined threshold value range may include a range of 0° to 1°. Further, the predefined threshold range value may depend upon the design of the at least one arm **202** of the angular sensor **112**.

[0043] FIG. 3 illustrates the proximity sensor **114** integrated within the at least one media holder **200** of the printer **102**, in accordance with an example embodiment of the present disclosure. FIG. 3 is described in conjunction with FIGS. 1-2B.

[0044] The proximity sensor **114** may be positioned at a base **300** of the printer **102** and underneath to the media **210**. In some embodiments, the proximity sensor **114** may correspond to at least one

of an electromagnetic filed sensor, capacitive proximity sensor, or a photoelectric proximity sensor. In some embodiments, the proximity sensor **114** may emit one or more rays. Further, the one or more rays may correspond to at least one of one or more electromagnetic fields or one or more electromagnetic radiation beams, for example, one or more infrared (IR) rays, ultrasonic sound waves (UV). The emitted one or more rays may hit the media **210** and may be reflected to the proximity sensor **114**. Further, the proximity sensor **114** may receive reflected one or more rays from the media **210**. In some embodiments, based on the received reflected one or more rays, a distance between the proximity sensor **114** and the media **210** may be determined. Further, there may be a predefined threshold value range of the distance, below or above which, improper media loading may be detected. In some embodiments, the predetermined threshold value range may depend on the design of the printer **102**. In one example embodiment, the predefined threshold value range of the printer **102** of a specific design may include a range of 2 centimeters (cm) to 5 cm. In some embodiments, the determined distance between the media **210** and the proximity sensor **114** below or above the predefined threshold value range may correspond to the improper loading of the media within the printer **102**.

[0045] In some embodiments, both the angular sensor **112** and the proximity sensor **114** may be used as a combination. The angular sensor **112** may ensure that the media **210** is placed at the at least one media holder **200** through the at least one arm **202**. Further, the proximity sensor **114** may ensure that the media may not be touching the base **300** of the printer **102**.

[0046] In some example embodiments, the proximity sensor **114** may correspond to the ultrasonic proximity sensor. Further, the ultrasonic proximity sensor may be configured to emit a high pitch sound. The emitted high pitch sound may hit the media **210** and reflect back to the ultrasonic proximity sensor. Further, a distance between the ultrasonic proximity sensor and the media **210** may be determined. The distance may be determined based at least on time taken by the high pitch sound to reflect back to the ultrasonic proximity sensor. Further, based at least on the determined distance, improper media loading may be detected. In other example embodiments, the proximity sensor **114** may correspond to at least one of a capacitive proximity sensor, a photoelectric proximity sensor, or any other proximity sensor known in the art.

[0047] In some example embodiments, the printer **102** may comprise at least two proximity sensors **114**. Both proximity sensors **114** may be placed at a predefined distance from each other to detect the distance between each proximity sensor **114** and the media **210**. Further, the distance between the proximity sensor **114** and the media **210** may be compared with the distance between at least one other proximity sensor **114** and the media **210**. Based on the comparison, any difference in the distance may indicate misloading of the media **210**.

[0048] FIG. 4 illustrates the display device **110** of the system **100**, in accordance with an example embodiment of the present disclosure. FIG. 4 is described in conjunction with FIGS. 1-3.

[0049] The display device **110** may include a user interface (UI) **400** for interacting with the user to display one or more notifications to the user. In some embodiments, the at least one processor **106** may generate one or more signals for the display device **110** to display the one or more notifications to the user. The at least one processor **106** may be configured to generate one or more signals upon determining the improper unloading of the media **210**. In some example embodiments, upon detecting the determined angle of the at least one arm **202** below or above the predefined threshold value range or distance between the media **210** and the proximity sensor **114** below or above the predefined threshold value, the UI **400** may display "IMPROPER LOADING OF MEDIA" to the user, as shown in FIG. 4.

[0050] Further, the UI **400** may receive the one or more commands from the user. Further, the UI **400** may provide an output to the user based on the received one or more commands. In one example embodiment, the user may provide a command to send a notification to a remote device such as a smartphone, when the improper loading of the media is detected. Thereafter, the UI **400** may display "NOTIFICATION SENT TO USER". In another example embodiment, the user may

provide a command to send a notification to determine the angle of the at least one arm **202** after a time interval of 30 seconds(s) and display the determined angle of the at least one arm **202** to the user. In yet another example embodiment, the user may provide a command to update the angle of the at least one arm **202** to the diameter of the media and display the diameter of the media to the user.

[0051] In some embodiments, the UI **400** may provide current date and time when the media **210** is loaded in the printer **102**. Further, the UI **400** may perform both the aforementioned actions simultaneously. The UI **400** may either be a Command Line Interface (CLI), Graphical User interface (GUI), or a voice interface. Further, the user may connect to the system **100** via the UI **400** using one or more communication protocols. The one or more communication protocols may include Bluetooth, 4G, 5G, Wi-Fi, or any other communication protocol known in the art.

[0052] FIG. **5** illustrates a flowchart showing steps of a method **500** of the system **100** to detect improper loading of media, in accordance with an example embodiment of the present disclosure. FIG. **6** is described in conjunction with FIGS. **1-4**.

[0053] At operation **502**, the information related to the angle of the at least one arm **202** that rotates in the predefined path **204** upon receiving the media **210** within the printer **102**, or the distance between the media **210** and the at least one media sensor **104** may be determined, via the at least one media sensor **104** installed within the printer **102**. In some embodiments, the at least one media sensor **104** may comprise at least one of the angular sensor **112** or the proximity sensor **114**.

[0054] In some embodiments, the angular sensor **112** may have the at least one arm **202**, and may be configured to determine the angle of the at least one arm **202**. In some embodiments, the proximity sensor **114** may emit one or more rays and may receive reflected one or more rays from the media **210** to determine the distance between the proximity sensor **114** and the media **210**. Further, the one or more rays may correspond to at least one of one or more electromagnetic fields or one or more electromagnetic radiation beams, for example, one or more infrared (IR) rays or ultrasonic sound waves (UV). In some embodiments, the predefined path **204** of the at least one arm **202** may define the first end **206** and the second end **208**. Further, the at least one arm **202** positioned at the first end **206** of the predefined path **204** may indicate an empty media condition and the at least one arm **202** positioned in the second end **208** of the predefined path **204** may indicate a full media condition. In some embodiments, the media **210** may push the at least one arm **202** towards the second end **208** of the predefined path **204** that may correspond to the proper loading of the media **210** within the printer **102**. In some embodiments, the determined information may be converted to diameter value of the media **210**.

[0055] For example, a user opens a cover of the printer **102** to load a paper. The at least one media sensor **104** determines information related to the angle of the at least one arm **202** that rotates in a predefined path **204** upon receiving the paper within the printer **102**, or the distance between the paper and the at least one media sensor **104**. The determined information obtained is 15° or 23 cm. The determined information is converted to diameter value of the paper.

[0056] At operation **504**, the determined information related to the angle or the distance between the media **210** and the at least one media sensor **104** may be received, via the at least one processor **106**, from the at least one media sensor **104**.

[0057] At operation **506**, the determined information may be compared, via the at least one processor **106**, with the predefined threshold value range associated with the angle of the at least one arm **202** or the distance between the media **210** and the at least one media sensor **104**. For example, the determined information of 15° or 23 cm is compared with a predefined threshold value range of 16-23° or 25 cm-30 cm.

[0058] At operation **508**, improper loading of the media **210** within the printer **102** may be determined, via the at least one processor **106**, based at least on the comparison. In some embodiments, the determined angle of the at least one arm **202** below or above the predefined threshold value range may correspond to the improper loading of the media **210** within the printer

102. In some embodiments, the determined distance between the media **210** and the proximity sensor **114** below or above the predefined threshold value range may correspond to the improper loading of the media **210** within the printer **102**. For example, based on the comparison since determined information of 15° or 23 cm is below the predefined threshold value range of 16-23° or 25 cm-30 cm, therefore improper loading of the media is determined.

[0059] In some embodiments, the at least one processor **106** generates one or more notifications upon determining the improper loading of the media **210** within the printer **102**. Further, the at least one display device **110** communicatively coupled to the at least one processor **106** may display the one or more notifications to the user.

[0060] In some embodiments, the system **100** may further generate, via the at least one processor **106**, one or more notifications upon determining the improper loading of the media **210** within the printer **102**. Further, the system **100** may display, via the at least one display device **110** the generated one or more notifications to the user. For example, “IMPROPER LOADING OF THE MEDIA” on the display device **110** when improper loading of the media is detected.

[0061] It will be appreciated that the method **500** may be implemented by the one or more embodiments disclosed herein, which may be combined or modified as desired or needed. Additionally, the steps in the method **500** may be modified, changed in order, performed differently, performed sequentially, concurrently or simultaneously, or otherwise modified as desired or needed.

[0062] In some embodiments, the system **100** and the method **500** prevents errors related to improper media loading by using proximity or media sensors. In some alternate embodiments, the introduction of the addition of the at least one media sensor further underscores its economic benefits, translating into substantial savings. In some embodiments, the system **100** may have a compact design that optimizes space utilization and enhances overall efficiency, making it a versatile and cost-effective solution for various applications. Additionally, the system **100** provide automated detection and display of media information to save time and resources by eliminating the need for manual configuration. Further, proper media loading detection by the system **100** and the method **500** minimizes the chances of misprints or errors, reducing material waste.

[0063] Embodiments may ensure that correct type and size of the media are loaded, preventing issues such as misalignment, smudging, or poor print quality. Embodiments may protect the printer from potential damage that may occur when improper media, such as excessively thick or incompatible materials, is loaded in the printer. Embodiments may provide real-time monitoring capability to ensure proper loading of the media inside the printer. Additionally, embodiments may be integrated into various types of printing equipment, such as laser printers, 3D printers, thermal printers, radio frequency identification (RFID) printers, etc., providing robust and reliable improper media loading detection solutions to improve efficiency, and maintain the longevity of the printing equipment.

[0064] Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe example embodiments in the context of certain example combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A system comprising: at least one media sensor installed within a printer, the at least one media sensor configured to determine information related to an angle of at least one arm that rotates in a predefined path upon receiving a media within the printer, or a distance between the media and the at least one media sensor; and at least one processor communicatively coupled to the at least one media sensor, the at least one processor configured to: receive the determined information related to the angle or the distance between the media and the at least one media sensor, from the at least one media sensor; compare the determined information with a predefined threshold value range associated with the angle of the at least one arm or the distance between the media and the at least one media sensor; and determine improper loading of the media within the printer based at least on the comparison.
2. The system of claim 1, wherein the at least one media sensor comprises at least one of an angular sensor or a proximity sensor.
3. The system of claim 2, wherein the angular sensor has the at least one arm, and is configured to determine the angle of the at least one arm.
4. The system of claim 2, wherein the proximity sensor emits one or more rays and receives reflected one or more rays from the media to determine the distance between the proximity sensor and the media, and wherein the one or more rays corresponds to at least one of one or more infrared waves (IR) or ultrasonic sound waves (UV).
5. The system of claim 3, wherein the at least one arm of the angular sensor is positioned along a surface of the media, and the proximity sensor is positioned at a base of the printer and underneath to the media.
6. The system of claim 1, wherein the predefined path of the at least one arm defines a first end and a second end, and wherein the at least one arm positioned at the first end of the predefined path indicates an empty media condition and the at least one arm positioned at the second end of the predefined path indicates a full media condition.
7. The system of claim 6, wherein the media pushes the at least one arm towards the second end of the predefined path that corresponds to the proper loading of the media within the printer.
8. The system of claim 1, wherein the determined angle of the at least one arm below or above the predefined threshold value range corresponds to the improper loading of the media within the printer.
9. The system of claim 4, wherein the determined distance between the media and the proximity sensor below or above the predefined threshold value range corresponds to the improper loading of the media within the printer.
10. The system of claim 1, wherein the at least one processor is configured to generate one or more notifications upon determining the improper loading of the media within the printer.
11. The system of claim 10, wherein the one or more notifications are displayed to a user over a display device communicatively coupled to the at least one processor.
12. A method comprising: determining, via at least one media sensor installed within a printer, information related to an angle of at least one arm that rotates in a predefined path upon receiving a media within the printer, or a distance between the media and the at least one media sensor; receiving, via at least one processor, the determined information related to the angle or the distance between the media and the at least one media sensor, from the at least one media sensor; comparing, via the at least one processor, the determined information with a predefined threshold value range associated with the angle of the at least one arm or the distance between the media and the at least one media sensor; and determining, via the at least one processor, improper loading of the media within the printer based at least on the comparison.
13. The method of claim 12, wherein the at least one media sensor comprises at least one of an

angular sensor or a proximity sensor.

14. The method of claim 13, wherein the angular sensor has the at least one arm, and is configured to determine the angle of the at least one arm.

15. The method of claim 13, wherein the proximity sensor emits one or more rays and receives reflected one or more rays from the media to determine the distance between the proximity sensor and the media, and wherein the one or more rays corresponds to at least one of one or more infrared waves (IR) or ultrasonic sound waves (UV).

16. The method of claim 12, wherein the predefined path of the at least one arm defines a first end and a second end, wherein the at least one arm positioned at the first end of the predefined path indicates an empty media condition and the at least one arm positioned in the second end of the predefined path indicates a full media condition.

17. The method of claim 16, wherein the media pushes the at least one arm towards the second end of the predefined path that corresponds to the proper loading of the media within the printer.

18. The method of claim 12, wherein the determined angle of the at least one arm below or above the predefined threshold value range corresponds to the improper loading of the media within the printer.

19. The method of claim 15, wherein the determined distance between the media and the proximity sensor below or above the predefined threshold value range, corresponds to the improper loading of the media within the printer.

20. The method of claim 12, further comprising: generating, via the at least one processor, one or more notifications upon determining the improper loading of the media within the printer, and displaying, via at least one display device communicatively coupled to the at least one processor, the one or more notifications to a user.
